

COMMUNICATION NETWORKS

ANSWER KEY AT THE BACK

40

90

45:00

1

ATTEMPT ONLINE

01

1 MARK

B Protocol

D Transmission medium

02

1 MARK

B Half-duplex

D Multiplex

03

1 MARK

B Only that one station is cut off

D Data starts circulating forever

04

1 MARK

B Five

D Eight

05

1 MARK

B 48 bits

D 128 bits

06

1 MARK

B Coaxial cable

D Shielded twisted pair

07 In bit-oriented framing, what is the flag byte that marks the start and end of a frame? 1 MARK
A 00000000 B 01111110
C 11111111 D 10101010

08 Which device operates at the physical layer and simply regenerates a signal without looking at addresses? 1 MARK
A Bridge B Router
C Repeater D Gateway

09 In CSMA/CD, what does the CD stand for? 1 MARK
A Collision Detection B Carrier Delay
C Channel Division D Collision Diversion

10 Which multiple-access method gives every station the whole bandwidth but only for its own turn in time? 1 MARK
A FDMA B TDMA
C CDMA D ALOHA

11 A message travels down the sending stack and each layer adds its own header. What is this process called, and what is its reverse at the receiver? 2 MARKS
A Multiplexing and demultiplexing B Encapsulation and decapsulation
C Framing and deframing D Segmentation and reassembly

12 Why can a MAC address not, by itself, deliver data to a host on a different network? 2 MARKS
A It is too short to hold a network number B It is a flat address with no network part, so a router cannot use it to choose a route
C It changes every time the machine is restarted D It is only readable by the manufacturer

13 In byte stuffing, what does the sender do when the data itself happens to contain a byte identical to the FLAG? 2 MARKS
A Deletes that byte B Inserts an extra ESC byte in front of it
C Replaces it with a zero byte D Sends the frame twice

14 In a Hamming code, where do the parity bits sit in the codeword? 2 MARKS
A At the end, all together B At the power-of-two positions 1, 2, 4, 8, ...
C At the odd positions D At the start, all together

15 For the same m-bit sequence number, how do the maximum sender window sizes of Go-Back-N and Selective Repeat compare? 2 MARKS

- A** Both are 2^{m-1}
- B** Go-Back-N is $2^m - 1$, Selective Repeat is 2^{m-1}
- C** Go-Back-N is 2^{m-1} , Selective Repeat is $2^m - 1$
- D** Both are 2^{m-1}

16 A station using 1-persistent CSMA finds the channel busy. What does it do? 2 MARKS

- A** Waits a random time, then senses again
- B** Keeps sensing continuously and transmits the instant the channel goes idle
- C** Gives up and reports failure
- D** Transmits anyway and relies on collision detection

17 What does a learning bridge use to build its forwarding table? 2 MARKS

- A** The destination address of every arriving frame **B** The source address of every arriving frame
C A configuration file written by the administrator **D** Broadcast queries sent to every port

18 In HDLC, what are the three frame types and what does each carry? 2 MARKS

- A** I-frames carry data, S-frames carry control and acknowledgements, U-frames manage the link
 - B** I-frames initialise, S-frames send, U-frames update
 - C** I-frames carry addresses, S-frames carry data, U-frames carry the FCS
 - D** All three carry data, but at different priorities

19 A network has an end-to-end path made of a 1 Gbps link, a 100 Mbps link and a 10 Mbps link. What is the maximum end-to-end throughput? 2 MARKS

- A** 1 Gbps **B** 370 Mbps (the average)
C 10 Mbps **D** 1.11 Gbps (the sum)

20 Which component of latency is the only one that varies wildly with how busy the network is? 2 MARKS

- A Propagation delay
 - B Transmission delay
 - C Processing delay
 - D Queuing delay

21 A pure ALOHA network transmits 100-bit frames on a shared 200 kbps channel. If the system produces 1000 frames per second, roughly how many get through? 3 MARKS

- A** 1000 **B** 368
C 184 **D** 500

22 What is the maximum throughput of pure ALOHA and of slotted ALOHA, and at what offered load does each occur? 3 MARKS

- A** 18.4% at $G = 0.5$ and 36.8% at $G = 1$ **B** 36.8% at $G = 1$ and 18.4% at $G = 0.5$
C 50% at $G = 1$ and 100% at $G = 2$ **D** 18.4% at $G = 1$ and 36.8% at $G = 0.5$

23 A CSMA/CD network runs at 10 Mbps and the maximum propagation time between two stations is 51.2 μ s. What is the minimum frame size? 3 MARKS

- A** 512 bits **B** 1024 bits
C 256 bits **D** 2048 bits

24 In a Stop-and-Wait system the line is 1 Mbps and one bit takes 20 ms for a round trip. Frames are 1000 bits. What is the link utilisation? 3 MARKS

- A** 50% **B** 20%
- C** 5% **D** 75%

25 The same link — 1 Mbps, 20 ms round trip, 1000-bit frames — now uses a protocol that can send 15 frames before stopping for an acknowledgement. What is the utilisation? 3 MARKS

- A** 15% **B** 75%
- C** 100% **D** 50%

26 Compute the CRC for the data word 1001 with the divisor 1011, and give the codeword that is transmitted. 3 MARKS

data : 1001
divisor : 1011 (so $r = 3$)

- A** CRC 110, codeword 1001110
B CRC 101, codeword 1001101
C CRC 011, codeword 1001011
D CRC 111, codeword 1001111

27 A 7-bit data word is to be protected by a Hamming code. How many parity bits are needed, and how long is the codeword? 3 MARKS

- A** 3 parity bits, 10-bit codeword **B** 4 parity bits, 11-bit codeword
C 4 parity bits, 12-bit codeword **D** 5 parity bits, 12-bit codeword

28 An 11-bit Hamming codeword arrives and the parity checks at positions 1 and 4 fail, while those at 2 and 8 pass. Which bit is wrong? 3 MARKS

- [illegible]

29 Apply bit stuffing to the sequence below, then say how many bits are actually transmitted. 3 MARKS

0 1 1 1 1 1 1 1 1 1 1 0

- A 13 bits, unchanged B 14 bits, one 0 inserted
C 15 bits, two 0s inserted D 16 bits, three 0s inserted

- 30** Why must a Selective Repeat receiver window be no larger than half the sequence-number space? 3 MARKS
- A To keep the buffers small
B So that a retransmitted old frame can never be mistaken for a new one after a wrap-around
C Because acknowledgements are numbered separately
D To make the modulo arithmetic easier
- 31** Four stations share a channel with CDMA. Station 1 has the Walsh code $[+1 +1 +1 +1]$ and station 2 has $[+1 -1 +1 -1]$. Station 1 sends bit 1 and station 2 sends bit 0, and no one else transmits. What does the channel carry? 3 MARKS
- station 1 : $[+1 +1 +1 +1]$ sends 1 $\rightarrow +1 \times$ code
station 2 : $[+1 -1 +1 -1]$ sends 0 $\rightarrow -1 \times$ code
- A $[+2 0 +2 0]$
B $[0 +2 0 +2]$
C $[+1 +1 +1 +1]$
D $[+2 +2 +2 +2]$
- 32** Six devices are to be connected in a full mesh. How many links are needed, and how many ports must each device have? 3 MARKS
- A 15 links, 5 ports each
B 30 links, 6 ports each
C 15 links, 6 ports each
D 36 links, 5 ports each
- 33** Two bridges are joined by two parallel links for redundancy, and a station sends one broadcast frame. What happens, and what fixes it? 3 MARKS
- A Nothing unusual; the second link is simply unused
B The frame circulates endlessly and multiplies, until the spanning tree algorithm blocks one port
C The bridges automatically load-balance across both links
D The frame is dropped because the destination is unknown
- 34** A frame arrives at a switch with the destination MAC address 01:00:5E:7F:00:01. How is it classified, and why? 3 MARKS
- A Unicast, because it is not all ones
B Multicast, because bit 0 of the first octet is 1
C Broadcast, because it starts with 01
D Unicast, because 01 is an even hexadecimal value
- 35** A source sends 5000 packets and 4750 arrive. Those 4750 packets carry 6,000,000 bits and the transfer took 4 seconds. What are the PDR and the throughput? 3 MARKS
- A PDR 95%, throughput 1.5 Mbps
B PDR 95%, throughput 1.2 Mbps
C PDR 5%, throughput 1.5 Mbps
D PDR 95%, throughput 6 Mbps
- 36** Three consecutive packets arrive with delays of 22 ms, 27 ms and 25 ms. What is the jitter between each consecutive pair? 3 MARKS
- A 5 ms and 2 ms
B 5 ms and -2 ms
C 2 ms and 3 ms
D 27 ms and 25 ms

- 37** Why does Wi-Fi use CSMA/CA rather than CSMA/CD? 3 MARKS
- A Radio is too fast for collision detection circuits B A wireless station cannot listen while it transmits, because its own signal drowns everything else out
- C Collisions never happen in wireless networks D CSMA/CD is patented and cannot be used in radio equipment
- 38** What is the vulnerable time of pure ALOHA and of slotted ALOHA, in terms of the frame time T_{fr} ? 3 MARKS
- A T_{fr} for pure, $2 \cdot T_{fr}$ for slotted B $2 \cdot T_{fr}$ for pure, T_{fr} for slotted
- C T_{fr} for both D $2 \cdot T_{fr}$ for both
- 39** In a Go-Back-N system with a window of 7, frames 0 to 6 are sent and frame 2 is lost. Frames 3, 4, 5 and 6 arrive undamaged. What does the sender retransmit? 3 MARKS
- A Only frame 2 B Frames 2, 3, 4, 5 and 6
- C Frames 0 to 6, all of them D Nothing, because the receiver buffers frames 3 to 6
- 40** Why is a single-mode fibre able to carry data much further than a multimode step-index fibre? 3 MARKS
- A Its core is made of a better grade of glass B Its core is so narrow that light travels essentially one path, so there is almost no modal dispersion
- C It uses a shorter wavelength of light D It has more cladding layers

The answer key

The letter, then why. Read the explanation even where you got it right — that is where most of the revision value is.

- 01 **C** The five are message, sender, receiver, transmission medium and protocol. Bandwidth is a property of the medium, not a component. Note also that the protocol is the only one of the five that is not physically located between sender and receiver — it is an agreement held at both ends.
- 02 **B** Half-duplex: data travels in both directions, but only one direction at a time. Simplex is one-way only (a keyboard, a monitor). Full-duplex is both directions at once (a telephone call).
- 03 **B** Every station has its own dedicated link to the central hub, so a broken cable isolates only that station. This is the star's main advantage over the bus, where a break in the backbone can bring down the whole network.
- 04 **C** Seven: physical, data link, network, transport, session, presentation, application — bottom to top. The TCP/IP suite collapses the top three into one and is usually drawn with five.
- 05 **B** 48 bits, normally written as six hexadecimal octets such as 00:1A:2B:3C:4D:5E. It is burned into the network interface card, which is why it is also called the physical or hardware address. An IPv4 address is the 32-bit one.
- 06 **C** Fibre carries light, not electricity, so no amount of electrical noise outside the cable can disturb it. Shielding helps a copper cable but never removes the problem completely.
- 07 **B** 01111110 — a 0, six 1s and a 0. Because that exact pattern must never appear inside the data, the sender uses bit stuffing: after five consecutive 1s in the payload it inserts a 0, and the receiver removes it again.
- 08 **C** A repeater receives a weakened signal and regenerates it. It has no idea what a frame or an address is. A hub is a multiport repeater. Bridges work at the data link layer, routers at the network layer, and a gateway can work at all layers.
- 09 **A** Carrier Sense Multiple Access with Collision Detection. The station listens before it sends, and keeps listening while it sends so it can abort as soon as a collision happens. Wireless stations cannot hear a collision while transmitting, which is why Wi-Fi uses CSMA/CA — collision avoidance — instead.
- 10 **B** TDMA divides the link in time: each station gets the full bandwidth during its slot. FDMA divides it in frequency, so each station gets a narrow band all the time. CDMA gives everybody all of both, separated by orthogonal codes.
- 11 **B** Encapsulation going down, decapsulation going up. Each layer treats whatever arrives from above as pure data and wraps its own header around it, so a layer never needs to understand the layers above it.
- 12 **B** A MAC address is flat and has no structure that says which network the host is on, so it is only meaningful inside one LAN. The IP address is hierarchical and carries a network part, which is what a router uses. On every hop the frame gets a fresh pair of MAC addresses while the IP addresses stay the same.
- 13 **B** An ESC byte is inserted before it, so the receiver knows the following byte is data rather than a delimiter and strips the ESC out. And if the data contains an ESC byte, that too must be preceded by an ESC — otherwise the receiver would lose track.
- 14 **B** Positions 1, 2, 4, 8, 16 ... and data bits fill everything else. That is what makes the scheme work: each parity bit checks the positions whose number contains its own bit, so adding up the positions of the failing checks gives the position of the wrong bit directly.
- 15 **B** Go-Back-N can use $2^m - 1$ because its receiver window is 1, so only one frame is ever expected. Selective Repeat buffers out-of-order frames, so its windows must not overlap after a wrap-around — which caps both windows at half the sequence space, 2^{m-1} .
- 16 **B** 1-persistent means it senses continuously and sends with probability 1 as soon as the line is free — which is exactly why it collides so often, since every waiting station jumps at the same instant. Non-persistent waits a random time before sensing again; p-persistent senses each slot and sends with probability p.

- 17 **B** The source address. If a frame carrying source address DD arrives on port P1, then DD must be reachable through P1. Entries are timed out and re-learned, so a station that is unplugged and moved to a different port is not stuck with a stale entry.
- 18 **A** Information frames carry user data (and can piggyback an acknowledgement), Supervisory frames carry acknowledgements and flow control but no data, and Unnumbered frames set up, manage and disconnect the link. The type is read from the first bits of the control field.
- 19 **C** The throughput of a path is set by its slowest link — the bottleneck. Upgrading the 1 Gbps link changes nothing at all. This is also why queues build up: data arrives at the bottleneck router faster than it can leave, and the wait is the queuing delay.
- 20 **D** Propagation depends only on distance and speed, transmission only on frame size and bandwidth, and processing only on the device. Queuing depends on traffic, so it is the component that makes a network feel unpredictable — and it is also what jitter measures the effect of.
- 21 **C** Frame time $T_{fr} = 100 \text{ bits} / 200 \text{ kbps} = 0.5 \text{ ms}$. Offered load $G = 1000 \times 0.5 \text{ ms} = 0.5$. Pure ALOHA throughput $S = G \cdot e^{-2G} = 0.5 \times e^{-1} = 0.184$. So $0.184 \times 1000 \approx 184$ frames per second succeed — the channel is being offered far more than it can carry. Drop the load to 200 frames/s and G falls to 0.1, giving $S = 0.0819$ and about 82 successes — fewer in total, but 41% of what was offered instead of 18%.
- 22 **A** Pure ALOHA: $S = G \cdot e^{-2G}$, which peaks at $S = 1/(2e) = 0.184$ when $G = 0.5$. Slotted ALOHA: $S = G \cdot e^{-G}$, peaking at $S = 1/e = 0.368$ when $G = 1$. Slotting halves the vulnerable time from $2 \cdot T_{fr}$ to T_{fr} , and that alone doubles the throughput.
- 23 **B** A station must still be transmitting when the worst-case collision signal gets back to it, so $T_{fr} \geq 2 \times T_p = 2 \times 51.2 \mu\text{s} = 102.4 \mu\text{s}$. Minimum frame = $10 \times 10^6 \times 102.4 \times 10^{-6} = 1024 \text{ bits}$, which is 128 bytes. (Standard Ethernet's famous 64-byte minimum comes from the same rule with $T_p = 25.6 \mu\text{s}$.)
- 24 **C** Bandwidth-delay product = $1 \times 10^6 \times 20 \times 10^{-3} = 20,000 \text{ bits}$ — the pipe holds 20,000 bits. Stop-and-Wait puts only one 1000-bit frame in flight per round trip, so utilisation = $1000 / 20,000 = 5\%$. The link spends 95% of its time idle, waiting for an acknowledgement.
- 25 **B** The pipe still holds 20,000 bits. Now $15 \times 1000 = 15,000 \text{ bits}$ are in flight per round trip, so utilisation = $15,000 / 20,000 = 75\%$. To reach 100% the window would have to be 20 frames — that is, the window must be at least the bandwidth-delay product divided by the frame size.
- 26 **A** The divisor is 4 bits so $r = 3$: append three zeros to get 1001000. Modulo-2 divide (XOR at each step): $1001 \oplus 1011 = 0010$, bring down 0 $\rightarrow 0100$, leading bit 0 so no XOR, bring down 0 $\rightarrow 1000$, $1000 \oplus 1011 = 0011$, bring down 0 $\rightarrow 0110$, leading bit 0, remainder = 110. Replace the three zeros with the remainder: codeword = 1001110. Check it by dividing 1001110 by 1011 — the remainder comes out 000.
- 27 **B** The condition is $2^r \geq m + r + 1$ with $m = 7$. Try $r = 3$: $8 \geq 11$ is false. Try $r = 4$: $16 \geq 12$ is true. So 4 parity bits, and the codeword is $7 + 4 = 11 \text{ bits}$, with the parity bits at positions 1, 2, 4 and 8.
- 28 **B** Add the position numbers of the failing checks: $1 + 4 = 5$. So bit 5 is the corrupted one, and flipping it repairs the codeword. This is the whole point of putting the parity bits at powers of two — the failing checks spell out the position in binary.
- 29 **C** There are eleven 1s in a row. After the first five 1s a 0 is stuffed; counting restarts, so after the next five 1s another 0 is stuffed. The result is 0 11111 0 11111 0 1 0 — two extra bits, so 15 are transmitted. The receiver removes any 0 that follows exactly five 1s, which is what stops the payload ever looking like the flag 01111110.
- 30 **B** The receiver buffers out-of-order frames, so at any moment it is willing to accept several sequence numbers. If the windows were more than half the space, an old retransmission could carry a number the receiver is currently expecting as *new*, and it would be accepted as fresh data — silent corruption. Capping both windows at 2^{m-1} makes the old and new ranges disjoint.

- 31 **B** A 1 is sent as +1 times the code, a 0 as -1 times the code. Station 1 contributes [+1 +1 +1 +1] and station 2 contributes [-1 +1 -1 +1]. Adding element by element gives [0 +2 0 +2]. Station 1 recovers its bit by multiplying by its own code and averaging: $(0-0 \dots) = (0 + 2 + 0 + 2)/4 = +1$, so bit 1. Station 2 gets $(0 - 2 + 0 - 2)/4 = -1$, so bit 0.
- 32 **A** Links = $n(n-1)/2 = 6 \times 5 / 2 = 15$, and each device needs $n - 1 = 5$ ports since it connects directly to every other device. The $n(n-1)/2$ comes from $n(n-1)$ ordered pairs divided by two, because a link between A and B is the same link as between B and A.
- 33 **B** A bridge floods a frame whose destination it does not know out of every port except the one it arrived on. With a loop present, each copy comes back, gets flooded again, and the number of copies grows without limit — a broadcast storm. The cure is the spanning tree algorithm: the bridges exchange BPDUs, elect a root, and block ports until exactly one path remains between any two points.
- 34 **B** Check for all ones first — FF:FF:FF:FF:FF:FF would be broadcast, and this is not. Then expand the first octet: 01 hex is 0000 0001, so bit 0 (the last bit written) is 1, which means multicast. Bit 1 would tell you whether the address is universally or locally administered. Note that 03 would also be multicast, even though it looks quite different in hex.
- 35 **A** $PDR = (4750 / 5000) \times 100 = 95\%$, so 5% of packets were lost. Throughput = $6,000,000 \text{ bits} / 4 \text{ s} = 1,500,000 \text{ bps} = 1.5 \text{ Mbps}$. Note that throughput is measured on what actually arrived, so the lost packets are already accounted for — and it is always lower than the link's bandwidth.
- 36 **A** Jitter = $|D(i+1) - D(i)|$. Between the first two: $|27 - 22| = 5 \text{ ms}$. Between the second and third: $|25 - 27| = 2 \text{ ms}$. It is an absolute value, so a delay that *drops* still counts as jitter — what matters for voice and video is that the spacing wobbled at all, not which way.
- 37 **B** The transmitted signal at the antenna is many orders of magnitude stronger than any signal arriving from a distant station, so the transmitter is deaf to collisions while it is sending. Since it cannot *detect* one, the protocol has to *avoid* one instead — with interframe spaces, a random back-off before sending, and an explicit acknowledgement, because silence cannot be taken as success. The hidden-terminal problem makes it worse still.
- 38 **B** In pure ALOHA a frame is destroyed by any transmission that starts in the Tfr before it or the Tfr during it, so the vulnerable window is $2 \cdot T_{fr}$. Slotting forces every station to start only at a slot boundary, which removes the partial overlap and cuts the window to T_{fr} . Halving the vulnerable time is exactly what doubles the peak throughput from 18.4% to 36.8%.
- 39 **B** A Go-Back-N receiver has a window of 1, so it discards anything out of order — frames 3 to 6 are thrown away even though they arrived perfectly. On timeout the sender goes back to frame 2 and resends everything from there. Selective Repeat would buffer 3 to 6 and resend only frame 2, which is the trade: Go-Back-N wastes bandwidth, Selective Repeat needs buffers and sorting at the receiver.
- 40 **B** In a multimode step-index fibre, rays enter at different angles and take paths of different lengths, so a pulse launched as a sharp spike arrives spread out — modal dispersion — and the spreading limits how close together pulses can be. A single-mode core is narrow enough that only one path is supported, so the pulse stays sharp. Graded-index fibre is the middle answer: the refractive index falls away from the centre, so the longer outer paths run faster and partly compensate.